

LECTURE NOTES ON INTEGRABLE SYSTEMS

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Week 1: Algebro-Geometric Solutions for the mKdV Equation (1)

2025-09-16

Objectives: Derive the mKdV equation from the associated spectral problem and construct the AKNS hierarchy by explicit recursion.

The modified Korteweg-de Vries (mKdV) equation is given by

$$(1) \quad u_t + 6u^2u_x + u_{xxx} = 0, \quad u(x, t) \in \mathbb{R}, \quad (x, t) \in \mathbb{R}^2.$$

A standard method for establishing integrability is via a Lax pair. We introduce a matrix function $\Phi(x, t)$ satisfying

$$\Phi_x = U\Phi, \quad \Phi_t = V\Phi,$$

where U and V are 2×2 matrices depending on u and its derivatives. The compatibility condition

$$\Phi_{xt} = \Phi_{tx} \iff U_t - V_x + [U, V] = 0,$$

with $[U, V] = UV - VU$, is the zero-curvature equation.

Following the AKNS formalism, the spatial part of the Lax pair is taken as

$$(2) \quad \Phi_x = U\Phi, \quad U = -i\lambda\sigma_3 + Q,$$

where

$$(3) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad \sigma_3 = \text{diag}(1, -1).$$

We introduce the transformation

$$(4) \quad \Phi(x; \lambda) = m(x; \lambda)e^{-i\lambda x\sigma_3}, \quad m(x; \lambda) \rightarrow \mathbb{I} \text{ as } \lambda \rightarrow \infty.$$

Substituting into the Lax equation gives

$$(5) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i\lambda \sigma_3 + Q.$$

Define

$$(6) \quad \Psi = m \sigma_3 m^{-1}.$$

Differentiating Ψ and using (5), we compute

$$\begin{aligned} \Psi_x &= m_x \sigma_3 m^{-1} - m \sigma_3 m^{-1} m_x m^{-1} \\ &= [m_x m^{-1}, m \sigma_3 m^{-1}] \\ &= [m_x m^{-1} - i\lambda m \sigma_3 m^{-1}, m \sigma_3 m^{-1}] \\ &= [U, \Psi]. \end{aligned}$$

Thus Ψ satisfies

$$(7) \quad \Psi_x = [U, \Psi],$$

which is the stationary zero-curvature condition.

The matrix $m(x; \lambda)$ admits the asymptotic expansion

$$(8) \quad m(x; \lambda) = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + \dots$$

From (6) and (8) we obtain the formal expansion

$$(9) \quad \Psi = \Psi_0 + \Psi_1 \lambda^{-1} + \Psi_2 \lambda^{-2} + \dots, \quad \Psi_0 = \sigma_3.$$

Substituting this into (7) yields

$$\begin{aligned} \sum_{j=1}^{\infty} \Psi_{j,x} \lambda^{-j} &= -i\lambda \left[\sigma_3, \sum_{j=1}^{\infty} \Psi_j \lambda^{-j} \right] + \left[Q, \sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right] \\ &= -i \sum_{j=0}^{\infty} [\sigma_3, \Psi_{j+1}] \lambda^{-j} + \sum_{j=0}^{\infty} [Q, \Psi_j] \lambda^{-j}. \end{aligned}$$

At order $O(1)$ we find

$$-i[\sigma_3, \Psi_1] + [Q, \sigma_3] = 0 \iff [i\Psi_1 + Q, \sigma_3] = 0,$$

which implies

$$\Psi_1^{\text{off}} = iQ.$$

For general $j \geq 1$, the coefficient of λ^{-j} gives

$$(10) \quad \Psi_{j,x} = -i[\sigma_3, \Psi_{j+1}] + [Q, \Psi_j] = -2i\sigma_3 \Psi_{j+1}^{\text{off}} + [Q, \Psi_j].$$

Hence

$$(11) \quad \Psi_{j,x}^{\text{off}} = -2i\sigma_3 \Psi_{j+1}^{\text{off}} + [Q, \Psi_j^{\text{diag}}] \iff \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3 (\Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}]),$$

and

$$(12) \quad \Psi_{j,x}^{\text{diag}} = [Q, \Psi_j^{\text{off}}] \iff \Psi_j^{\text{diag}} = \partial_x^{-1} ([Q, \Psi_j^{\text{off}}]).$$

Thus, the recursion relation (11) provides a direct way to compute the off-diagonal coefficients algebraically in terms of lower-order terms, while the diagonal recursion (12) requires performing an integration. In practice, this makes the off-diagonal part more convenient to obtain explicitly, whereas the diagonal part is generally harder to compute in closed form.

By (6), the additional constraint $\Psi^2 = \mathbb{I}$ provides algebraic relations among the coefficients. Substituting the expansion (9) yields

$$\mathbb{I} = \left(\sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right)^2 = \sum_{j=0}^{\infty} \left(\sum_{k=0}^j \Psi_k \Psi_{j-k} \right) \lambda^{-j},$$

so that

$$(13) \quad \sum_{k=0}^j \Psi_k \Psi_{j-k} = 0, \quad j \geq 1.$$

Separating diagonal and off-diagonal parts, for $j \geq 1$ we obtain

$$(14) \quad \sum_{k=0}^j (\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}}) = 0,$$

$$(15) \quad \sum_{k=0}^j (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Since $\Psi_0 = \sigma_3$, the off-diagonal constraint (15) for $j \geq 2$ takes the form

$$\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 + \sum_{k=1}^{j-1} (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Noting that $\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 = 0$, this simplifies to

$$(16) \quad \sum_{k=1}^{j-1} (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

It should be noted that from the off-diagonal constraint (15) one can only deduce the reduced form (16), which merely expresses a compatibility condition between lower-order coefficients. In particular, this relation does not provide a practical means to compute the diagonal parts explicitly.

For the diagonal part (14) we have

$$\sigma_3 \Psi_j^{\text{diag}} + \Psi_j^{\text{diag}} \sigma_3 + \sum_{k=1}^{j-1} (\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}}) = 0, \quad j \geq 2.$$

Since $\sigma_3 \Psi_j^{\text{diag}} = \Psi_j^{\text{diag}} \sigma_3$, this simplifies to

$$(17) \quad \Psi_j^{\text{diag}} = -\frac{1}{2} \sigma_3 \sum_{k=1}^{j-1} (\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}}), \quad j \geq 2.$$

We will use this reduced relation (17) to compute the diagonal part Ψ_j^{diag} recursively at each order. This provides an explicit algebraic formula for the diagonal coefficients in terms of lower-order diagonal and off-diagonal terms.

For $j = 1$, the condition (13) reads

$$\sigma_3 \Psi_1 + \Psi_1 \sigma_3 = 2\sigma_3 \Psi_1^{\text{diag}} = 0,$$

so that $\Psi_1 = iQ$.

Using (11) and (17) recursively, we find

$$\Psi_2^{\text{off}} = -\frac{1}{2} \sigma_3 Q_x, \quad \Psi_2^{\text{diag}} = \frac{1}{2} \sigma_3 Q^2,$$

and hence

$$\Psi_2 = \frac{1}{2} \sigma_3 (Q^2 - Q_x).$$

For the next coefficient, we compute

$$\begin{aligned} \Psi_3^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{2,x}^{\text{off}} - [Q, \Psi_2^{\text{diag}}]) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{1}{2} \sigma_3 Q_{xx} - [Q, \frac{1}{2} \sigma_3 Q^2] \right) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{1}{2} \sigma_3 Q_{xx} - \frac{1}{2} (Q \sigma_3 Q^2 - \sigma_3 Q^2 Q) \right) \\ &= -\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3. \end{aligned}$$

For the diagonal part,

$$\begin{aligned} \Psi_3^{\text{diag}} &= -\frac{1}{2} \sigma_3 \left(\Psi_1^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{off}} \Psi_1^{\text{off}} + \Psi_1^{\text{diag}} \Psi_2^{\text{diag}} + \Psi_2^{\text{diag}} \Psi_1^{\text{diag}} \right) \\ &= -\frac{1}{2} \sigma_3 \left((iQ) \left(-\frac{1}{2} \sigma_3 Q_x \right) + \left(-\frac{1}{2} \sigma_3 Q_x \right) (iQ) \right) \\ &= -\frac{1}{2} \sigma_3 \left(-\frac{i}{2} (Q \sigma_3 Q_x + \sigma_3 Q_x Q) \right) \\ &= \frac{i}{4} (Q_x Q - Q Q_x) = 0. \end{aligned}$$

Proceeding to Ψ_4 , the off-diagonal recursion (11) with $\Psi_3^{\text{diag}} = 0$ gives

$$\begin{aligned} \Psi_4^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{3,x}^{\text{off}} - [Q, \Psi_3^{\text{diag}}]) = \frac{i}{2} \sigma_3 \Psi_{3,x}^{\text{off}} \\ &= \frac{i}{2} \sigma_3 \partial_x \left(-\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3 \right) \\ &= \frac{i}{2} \sigma_3 \left(-\frac{i}{4} Q_{xxx} + \frac{i}{2} (Q^3)_x \right) \\ &= \frac{1}{8} \sigma_3 Q_{xxx} - \frac{1}{4} \sigma_3 (Q^3)_x. \end{aligned}$$

Since $Q_x Q = Q Q_x$ for the present Q given in (3), we have

$$(Q^3)_x = Q_x Q^2 + Q Q_x Q + Q^2 Q_x = 3Q_x Q^2,$$

and therefore

$$\Psi_4^{\text{off}} = \frac{1}{8} \sigma_3 Q_{xxx} - \frac{3}{4} \sigma_3 Q_x Q^2.$$

For the diagonal part, only the nonvanishing combinations contribute; terms involving Ψ_1^{diag} or Ψ_3^{diag} are zero and omitted. Hence

$$\begin{aligned}\Psi_4^{\text{diag}} &= -\frac{1}{2}\sigma_3\left(\Psi_1^{\text{off}}\Psi_3^{\text{off}} + \Psi_3^{\text{off}}\Psi_1^{\text{off}} + \Psi_2^{\text{off}}\Psi_2^{\text{off}} + \Psi_2^{\text{diag}}\Psi_2^{\text{diag}}\right) \\ &= -\frac{1}{2}\sigma_3\left((iQ)\left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3\right) + \left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3\right)(iQ)\right. \\ &\quad \left.+ \left(-\frac{1}{2}\sigma_3Q_x\right)^2 + \left(\frac{1}{2}\sigma_3Q^2\right)^2\right) \\ &= -\frac{1}{8}\sigma_3(QQ_{xx} + Q_{xx}Q - 3Q^4 - Q_x^2).\end{aligned}$$

Since $(-i\lambda\Psi)_+ = -i\lambda\sigma_3 + Q$, we obtain

$$\Phi_x\Phi^{-1} = U = (-i\lambda\Psi)_+.$$

Here the subscript $(\cdot)_+$ denotes the projection onto the nonnegative powers of λ in the Laurent expansion. This motivates the definition of the AKNS hierarchy by prescribing the time evolution

$$(18) \quad \Phi_{t_n} = (-i\lambda^n\Psi)_+\Phi \equiv V_n\Phi.$$

The first few members of the hierarchy are given by

$$\begin{aligned}n = 2 &\Rightarrow \text{nonlinear Schrödinger (NLS) flow,} \\ n = 3 &\Rightarrow \text{complex mKdV flow,} \Rightarrow \text{mKdV flow.}\end{aligned}$$

In terms of the expansion coefficients Ψ_j , the operators V_n take the form

$$\begin{aligned}V_2 &= -i(\lambda^2\Psi_0 + \lambda\Psi_1 + \Psi_2), \\ V_3 &= -i(\lambda^3\Psi_0 + \lambda^2\Psi_1 + \lambda\Psi_2 + \Psi_3).\end{aligned}$$

The zero-curvature condition

$$U_t - V_x + [U, V] = 0,$$

with $V_{\text{mKdV}} = 4V_3$, yields at order $O(1)$

$$Q_t + 4i\Psi_{3,x} + [Q, -4i\Psi_3] = 0.$$

Taking the off-diagonal part gives

$$Q_t + 4i\Psi_{3,x}^{\text{off}} = 0,$$

that is,

$$(19) \quad Q_t + 4i\left(-\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3\right)_x = Q_t + Q_{xxx} - 2(Q^3)_x = 0.$$

Since

$$Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad Q^2 = -u^2\mathbb{I},$$

this reduces to the scalar equation

$$u_t + u_{xxx} + 2(u^3)_x = 0,$$

which is precisely the mKdV equation (1).

Using the definitions (4) and (6) we have

$$(20) \quad \Psi = \Phi\sigma_3\Phi^{-1}.$$

Differentiating Ψ with respect to t_n and using (18) gives

$$\begin{aligned}\Psi_{t_n} &= \Phi_{t_n}\sigma_3\Phi^{-1} - \Phi\sigma_3\Phi^{-1}\Phi_{t_n}\Phi^{-1} \\ &= [\Phi_{t_n}\Phi^{-1}, \Psi] = [V_n, \Psi].\end{aligned}$$

Thus

$$(21) \quad \Psi_{t_n} = [V_n, \Psi].$$

Next we simplify the zero-curvature condition for the pair (U, V_n) . The zero-curvature equation is

$$U_{t_n} - (V_n)_x + [U, V_n] = 0,$$

and since $U = -i\lambda\sigma_3 + Q$ we have $U_{t_n} = Q_{t_n}$. With the normalization

$$V_n = \alpha_n (-i\lambda^n \Psi)_+$$

we obtain

$$Q_{t_n} = (V_n)_x - [U, V_n] = \alpha_n (-i\lambda^n \Psi_x)_+ - [U, \alpha_n (-i\lambda^n \Psi)_+].$$

Pulling out constants and using $\Psi_x = [U, \Psi]$ gives

$$(22) \quad Q_{t_n} = -i\alpha_n \left((\lambda^n [U, \Psi])_+ - [U, (\lambda^n \Psi)_+] \right).$$

Write $U = -i\lambda\sigma_3 + Q$ and split the first term in (22):

$$\lambda^n [U, \Psi] = -i\lambda^{n+1} [\sigma_3, \Psi] + \lambda^n [Q, \Psi].$$

Because Q does not depend on λ we have

$$(\lambda^n [Q, \Psi])_+ = [Q, (\lambda^n \Psi)_+],$$

so the Q -commutator contributions cancel in (22). The remaining terms therefore come only from the σ_3 -commutator:

$$\begin{aligned} Q_{t_n} &= -i\alpha_n \left(-i(\lambda^{n+1} [\sigma_3, \Psi])_+ + i\lambda [\sigma_3, (\lambda^n \Psi)_+] \right) \\ &= \alpha_n \left(\lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ \right). \end{aligned}$$

To simplify the difference of finite sums, use the expansions

$$\Psi = \sum_{j \geq 0} \Psi_j \lambda^{-j}, \quad (\lambda^n \Psi)_+ = \sum_{j=0}^n \lambda^{n-j} \Psi_j.$$

A direct comparison of coefficients shows

$$\begin{aligned} \lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ &= \sum_{j=0}^n \lambda^{n+1-j} [\sigma_3, \Psi_j] - \sum_{j=0}^{n+1} \lambda^{n+1-j} [\sigma_3, \Psi_j] \\ &= -\lambda^{n+1-(n+1)} [\sigma_3, \Psi_{n+1}] \\ &= -[\sigma_3, \Psi_{n+1}]. \end{aligned}$$

Hence the evolution of Q is

$$(23) \quad Q_{t_n} = -\alpha_n [\sigma_3, \Psi_{n+1}].$$

Since diagonal parts commute with σ_3 and the commutator doubles the off-diagonal part, we may equivalently write

$$(24) \quad Q_{t_n} = -2\alpha_n \sigma_3 \Psi_{n+1}^{\text{off}}.$$

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